

SC2107/CE2107 Microprocessor System Design and Development

Tutorial 6

Serial Bus and Display

1. Discuss the common and different characteristics of SPI and I2C?

Serial, master device controls CLK, usually 1 master at a time.

Difference - SPI 3/4 wires, I2C only 2 (SDA, SCLK), SPI usually faster throughput, SPI slave chip select, I2C via address

2. In the I2C protocol, in order to read from the slave device, you must first tell it which of its internal addresses you want to read from. So a read of the slave actually starts off by writing to it. Let's look at the sequence of communicating with a peripheral module.
 - Send the Start Sequence
 - Send 0xC0 (I2C address of the peripheral module with R/W set to low)
 - Send 0x01 (Internal address of the register that we want to read from)
 - Send a Start Sequence again
 - Send 0xC1 (I2C address of the peripheral module with R/W set to high)
 - Read a byte from the peripheral device
 - Send the Stop Sequence

Illustrate the SCL and SDA lines of the I2C bus to reflect the above sequence.

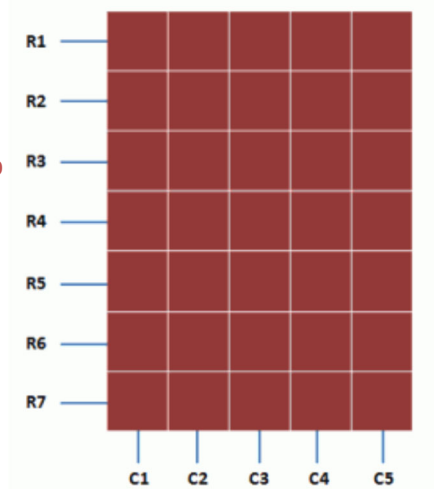
3. In the I2C protocol, there could be a situation where the slave is not ready to send data.

Why is this a problem and describe the solution in the I2C protocol to overcome this problem.

Clock stretching, instead of the master controlling the clk, slave will pull the clk down. Slave then gets the data and places it in the transmission register and releases the clock line, allowing the pull-up resistor to finally pull it high. From the masters point of view, it will issue the first clk pulse of the read by making SCLK high then check to see if it really has gone high.

4. A matrix LED as shown in the figure below is to be interfaced to a microprocessor. Explain how the interfacing will be done and the factors that should be considered.

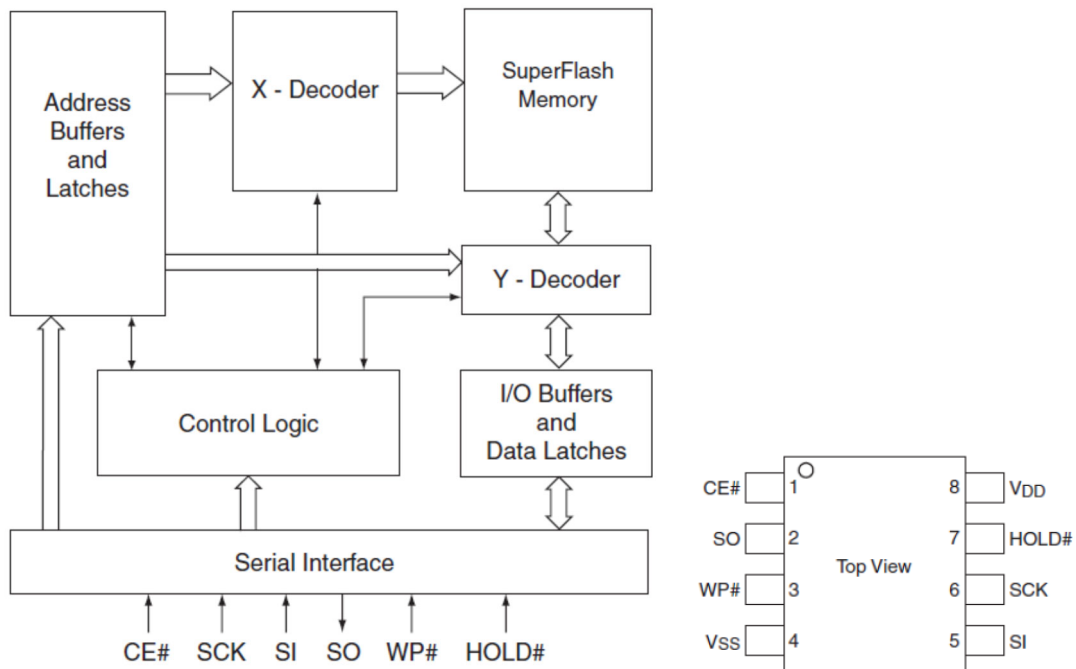
Whether Port C is able to sink 7 LED worth of current, if all R1 -R7 is lit. Usually a external device to help sink the current, device is called current-sinking transistor array. Port C tells transistor array which port to sink and it will do so accordingly.



Optional

5. The SPI is a common interface for a microprocessor to communicate with peripheral chips. In order to communicate effectively, the programmer must have a good understanding of the protocols associated with that particular chip. These protocols may be unique to that particular chip or may be generalized due to an industry standard.

We will look at some of the protocols of the 32Mbit SPI Flash chip (SST25VF032B) to have a better understanding.

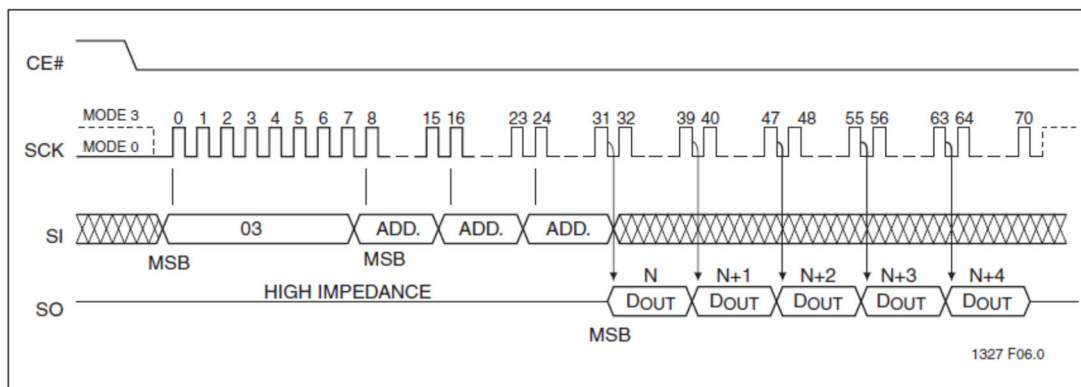


Block Diagram and Pin-Out of the SPI Flash Chip

The Device Operation Protocol is as follows:

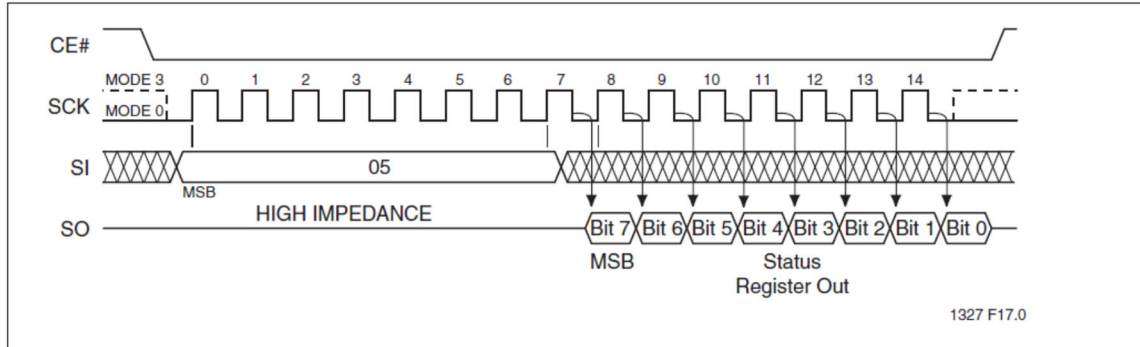
Instruction	Description	Op Code Cycle ¹	Address Cycle(s) ²	Dummy Cycle(s)	Data Cycle(s)	Maximum Frequency
Read	Read Memory	0000 0011b (03H)	3	0	1 to ∞	25 MHz
High-Speed Read	Read Memory at higher speed	0000 1011b (0BH)	3	1	1 to ∞	80 MHz
4 KByte Sector-Erase ³	Erase 4 KByte of memory array	0010 0000b (20H)	3	0	0	80 MHz
32 KByte Block-Erase ⁴	Erase 32KByte block of memory array	0101 0010b (52H)	3	0	0	80 MHz
64 KByte Block-Erase ⁵	Erase 64 KByte block of memory array	1101 1000b (D8H)	3	0	0	80 MHz
Chip-Erase	Erase Full Memory Array	0110 0000b (60H) or 1100 0111b (C7H)	0	0	0	80 MHz
Byte-Program	To Program One Data Byte	0000 0010b (02H)	3	0	1	80 MHz
AAI-Word-Program ⁶	Auto Address Increment Programming	1010 1101b (ADH)	3	0	2 to ∞	80 MHz
RDSR ⁷	Read-Status-Register	0000 0101b (05H)	0	0	1 to ∞	80 MHz
EWSR	Enable-Write-Status-Register	0101 0000b (50H)	0	0	0	80 MHz
WRSR	Write-Status-Register	0000 0001b (01H)	0	0	1	80 MHz
WREN	Write-Enable	0000 0110b (06H)	0	0	0	80 MHz
WRDI	Write-Disable	0000 0100b (04H)	0	0	0	80 MHz
RDID ⁸	Read-ID	1001 0000b (90H) or 1010 1011b (ABH)	3	0	1 to ∞	80 MHz
JEDEC-ID	JEDEC ID read	1001 1111b (9FH)	0	0	3 to ∞	80 MHz

The following figure shows the waveform for the 'Read Sequence' command.



- Decode the protocol sequence from the microprocessor's perspective.

The following figure shows another SPI transaction with the device.



- Decode the protocol command.
- Decode the protocol sequence from the microprocessor's perspective.